Second, if x and y are numbers in P, then both xy and $x + y$ are in P.

In the real number system the set P is the set of *positive* numbers. In the real number system we say x is greater than y, written $x > y$, if and only if $x - y$ is in P. Discuss why the complex number system has no such subset P. [*Hint*: Consider i and $-i$.]

Computer Lab Assignments

In Problems 56–59, use a CAS* to write the given complex number in the form $a + bi$. In *Mathematica* use the command **ComplexExpand**.

56. $2 + i - \dfrac{1}{3 - i}$ **57.** $\dfrac{5i}{2 + 3i}$

58. $\dfrac{1 + i}{2 - 5i}$ **59.** $\dfrac{(1 + 3i)(2 + i)}{3 - i}$

1.2 Complex Plane

INTRODUCTION A complex number $z = x + iy$ is uniquely determined by an *ordered pair* of real numbers (x, y). The first and second entries of the ordered pairs correspond, in turn, with the real and imaginary parts of the complex number. For example, the ordered pair $(2, -3)$ corresponds to the complex number $z = 2 - 3i$. Conversely, $z = 2 - 3i$ determines the ordered pair $(2, -3)$. The numbers 7, i, and $-5i$ are equivalent to $(7, 0)$, $(0, 1)$, $(0, -5)$, respectively. In this manner we are able to associate a complex number $z = x + iy$ with a *point* (x, y) in a coordinate plane.

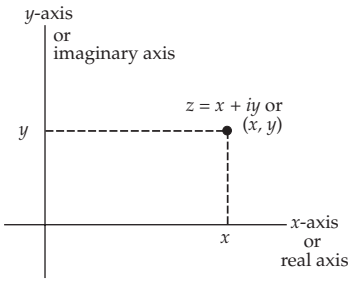


FIGURE 1.2.1 z-plane

Complex Plane Because of the correspondence between a complex number $z = x + iy$ and one and only one point (x, y) in a coordinate plane, we shall use the terms *complex number* and *point* interchangeably. The coordinate plane illustrated in FIGURE 1.2.1 is called the **complex plane** or simply the **z-plane**. The horizontal or x-axis is called the **real axis** because each point on that axis represents a real number. The vertical or y-axis is called the **imaginary axis** because a point on that axis represents a pure imaginary number.

Vectors In other courses you have undoubtedly seen that the numbers in an ordered pair of real numbers can be interpreted as the components of a vector. Thus, a complex number $z = x + iy$ can also be viewed as a two-dimensional position **vector**, that is, a vector whose initial point is the origin and whose terminal point is the point (x, y). See FIGURE 1.2.2. This vector interpretation prompts us to define the length of the vector z as the distance $\sqrt{x^2 + y^2}$ from the origin to the point (x, y). This length is given a special name.

A brief review of vectors can be found in Section 1.2 of the student study guide. ▶

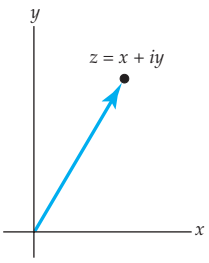


FIGURE 1.2.2 z as a position vector

DEFINITION 1.2.1 Modulus

The **modulus** of a complex number $z = x + iy$ is the real number

$$|z| = \sqrt{x^2 + y^2}. \tag{1}$$

The modulus $|z|$ of a complex number z is also called the **absolute value** of z. We shall use both words *modulus* and *absolute value* throughout this text.

EXAMPLE 1 Modulus of a Complex Number

If $z = 2 - 3i$, then from (1) we find the modulus of the number to be $|z| = \sqrt{2^2 + (-3)^2} = \sqrt{13}$. If $z = -9i$, then (1) gives $|-9i| = \sqrt{(-9)^2} = 9$. ☰

*Throughout this text we shall use the abbreviation CAS for "computer algebra system."

Properties Recall from (4) of Section 1.1 that for any complex number $z = x + iy$ the product $z\bar{z}$ is a real number; specifically, $z\bar{z}$ is the sum of the squares of the real and imaginary parts of z: $z\bar{z} = x^2 + y^2$. Inspection of (1) then shows that $|z|^2 = x^2 + y^2$. The relations

$$|z|^2 = z\bar{z} \quad \text{and} \quad |z| = \sqrt{z\bar{z}} \tag{2}$$

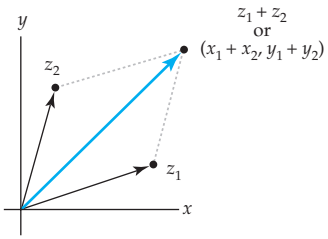
deserve to be stored in memory. The modulus of a complex number z has the additional properties.

$$|z_1 z_2| = |z_1|\,|z_2| \quad \text{and} \quad \left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}. \tag{3}$$

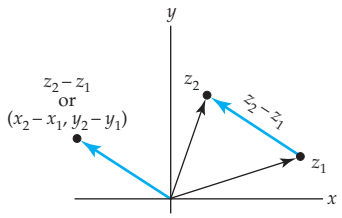
Note that when $z_1 = z_2 = z$, the first property in (3) shows that

$$|z^2| = |z|^2. \tag{4}$$

The property $|z_1 z_2| = |z_1|\,|z_2|$ can be proved using (2) and is left as an exercise. See Problem 49 in Exercises 1.2.



(a) Vector sum



(b) Vector difference

FIGURE 1.2.3 Sum and difference of vectors

Distance Again The addition of complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ given in Section 1.1, when stated in terms of ordered pairs:

$$(x_1,\ y_1) + (x_2,\ y_2) = (x_1 + x_2,\ y_1 + y_2)$$

is simply the component definition of vector addition. The vector interpretation of the sum $z_1 + z_2$ is the vector shown in **FIGURE 1.2.3(a)** as the main diagonal of a parallelogram whose initial point is the origin and terminal point is $(x_1 + x_2,\ y_1 + y_2)$. The difference $z_2 - z_1$ can be drawn either starting from the terminal point of z_1 and ending at the terminal point of z_2, or as a position vector whose initial point is the origin and terminal point is $(x_2 - x_1,\ y_2 - y_1)$. See Figure 1.2.3(b). In the case $z = z_2 - z_1$, it follows from (1) and Figure 1.2.3(b) that the **distance between two points** $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ in the complex plane is the same as the distance between the origin and the point $(x_2 - x_1,\ y_2 - y_1)$; that is, $|z| = |z_2 - z_1| = |(x_2 - x_1) + i(y_2 - y_1)|$ or

$$|z_2 - z_1| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}. \tag{5}$$

When $z_1 = 0$, we see again that the modulus $|z_2|$ represents the distance between the origin and the point z_2.

EXAMPLE 2 Set of Points in the Complex Plane

Describe the set of points z in the complex plane that satisfy $|z| = |z - i|$.

Solution We can interpret the given equation as equality of distances: The distance from a point z to the origin equals the distance from z to the point i. Geometrically, it seems plausible from **FIGURE 1.2.4** that the set of points z lie on a horizontal line. To establish this analytically, we use (1) and (5) to write $|z| = |z - i|$ as:

$$\sqrt{x^2 + y^2} = \sqrt{x^2 + (y - 1)^2}$$
$$x^2 + y^2 = x^2 + (y - 1)^2$$
$$x^2 + y^2 = x^2 + y^2 - 2y + 1.$$

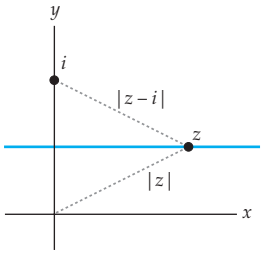


FIGURE 1.2.4 Horizontal line is the set of points satisfying $|z| = |z - i|$

The last equation yields $y = \frac{1}{2}$. Since the equality is true for arbitrary x, $y = \frac{1}{2}$ is an equation of the horizontal line shown in color in Figure 1.2.4. Complex numbers satisfying $|z| = |z - i|$ can then be written as $z = x + \frac{1}{2}i$. ☰

Inequalities In the *Remarks* at the end of the last section we pointed out that no order relation can be defined on the system of complex numbers. However, since $|z|$ is a real number, we can compare the absolute values of two complex numbers. For example, if $z_1 = 3 + 4i$ and $z_2 = 5 - i$, then $|z_1| = \sqrt{25} = 5$ and $|z_2| = \sqrt{26}$ and, consequently, $|z_1| < |z_2|$. In view of (1), a geometric interpretation of the last inequality is simple: *The point* (3, 4) *is closer to the origin than the point* (5, −1).

Now consider the triangle given in **FIGURE 1.2.5** with vertices at the origin, z_1, and $z_1 + z_2$. We know from geometry that the length of the side of the triangle corresponding to the vector $z_1 + z_2$ cannot be longer than the sum of the lengths of the remaining two sides. In symbols we can express this observation by the inequality

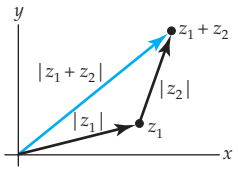


FIGURE 1.2.5 Triangle with vector sides

$$|z_1 + z_2| \le |z_1| + |z_2|. \tag{6}$$

This inequality can be derived using the properties of complex numbers in Section 1.1. See Problem 50 in Exercises 1.2. ▶

The result in (6) is known as the **triangle inequality**. Now from the identity $z_1 = z_1 + z_2 + (-z_2)$, (6) gives

$$|z_1| = |z_1 + z_2 + (-z_2)| \le |z_1 + z_2| + |-z_2|\,.$$

Since $|z_2| = |-z_2|$ (see Problem 47 in Exercises 1.2), solving the last result for $|z_1 + z_2|$ yields another important inequality:

$$|z_1| - |z_2| \le |z_1 + z_2|. \tag{7}$$

But because $z_1 + z_2 = z_2 + z_1$, (7) can be written in the alternative form $|z_1 + z_2| = |z_2 + z_1| \ge |z_2| - |z_1| = -(|z_1| - |z_2|)$ and so combined with the last result implies

$$\big|\,|z_1| - |z_2|\,\big| \le |z_1 + z_2|. \tag{8}$$

It also follows from (6) by replacing z_2 by $-z_2$ that $|z_1 + (-z_2)| \le |z_1| + |(-z_2)| = |z_1| + |z_2|$. This result is the same as

$$|z_1 - z_2| \le |z_1| + |z_2|\,. \tag{9}$$

From (8) with z_2 replaced by $-z_2$, we also find

$$\big|\,|z_1| - |z_2|\,\big| \le |z_1 - z_2|. \tag{10}$$

In conclusion, we note that the triangle inequality (6) extends to any finite sum of complex numbers:

$$|z_1 + z_2 + z_3 + \cdots + z_n| \le |z_1| + |z_2| + |z_3| + \cdots + |z_n|\,. \tag{11}$$

The inequalities (6), (8), and (10) will be important when we work with integrals in Chapters 5 and 6.

EXAMPLE 3 An Upper Bound

Find an upper bound for $\left|\dfrac{-1}{z^4+3z^2+2}\right|$ if $|z|=2$.

Solution By the second result in (3), the absolute value of a quotient is the quotient of the absolute values. Thus with $|-1|=1$, we want to find a positive real number M such that

$$\frac{1}{|z^4+3z^2+2|} \le M.$$

To accomplish this task we want the denominator as small as possible. Since $z^4+3z^2+2=(z^2+1)(z^2+2)$ we can use the first result in (3) and (8) to write

$$|z^4+3z^2+2| = |z^2+1|\cdot|z^2+2| \ge \big||z^2|-1\big|\cdot\big||z^2|-2\big|. \tag{12}$$

Using $|z|=2$, (12) becomes

$$\begin{aligned} |z^4+3z^2+2| \ge \big||z^2|-1\big|\cdot\big||z^2|-2\big| &= \big||z|^2-1\big|\cdot\big||z|^2-2\big| \\ &= |4-1|\cdot|4-2| \\ &= 6. \end{aligned}$$

Hence for $|z|=2$ we have

$$\left|\frac{-1}{z^4+3z^2+2}\right| = \frac{1}{|z^4+3z^2+2|} \le \frac{1}{6}.$$

☰

Remarks

We have seen that the triangle inequality $|z_1+z_2| \le |z_1|+|z_2|$ indicates that the length of the vector z_1+z_2 cannot exceed the sum of the lengths of the individual vectors z_1 and z_2. But the results given in (3) are interesting. The product z_1z_2 and quotient z_1/z_2, $(z_2 \neq 0)$, are complex numbers and so are vectors in the complex plane. The equalities $|z_1z_2| = |z_1|\,|z_2|$ and $|z_1/z_2| = |z_1|/|z_2|$ indicate that the lengths of the vectors z_1z_2 and z_1/z_2 are exactly equal to the product of the lengths and to the quotient of the lengths, respectively, of the individual vectors z_1 and z_2.

EXERCISES 1.2

Answers to selected odd-numbered problems begin on page ANS-2.

In Problems 1–4, interpret z_1 and z_2 as vectors. Graph z_1, z_2, and the indicated sum and difference as vectors.

1. $z_1 = 4+2i,\ z_2 = -2+5i;\ z_1+z_2,\ z_1-z_2$
2. $z_1 = 1-i,\ z_2 = 1+i;\ z_1+z_2,\ z_1-z_2$
3. $z_1 = 5+4i,\ z_2 = -3i;\ 3z_1+5z_2,\ z_1-2z_2$
4. $z_1 = 4-3i,\ z_2 = -2+3i;\ 2z_1+4z_2,\ z_1-z_2$
5. Given that $z_1 = 5-2i$ and $z_2 = -1-i$, find a vector z_3 in the same direction as z_1+z_2 but four times as long.

6. (**a**) Plot the points $z_1 = -2 - 8i,\ z_2 = 3i,\ z_3 = -6 - 5i$.

(**b**) The points in part (a) determine a triangle with vertices at z_1, z_2, and z_3, respectively. Express each side of the triangle as a difference of vectors.

7. In Problem 6, determine whether the points z_1, z_2, and z_3 are the vertices of a right triangle.

8. The three points $z_1 = 1 + 5i,\ z_2 = -4 - i,\ z_3 = 3 + i$ are vertices of a triangle. Find the length of the median from z_1 to the side $z_3 - z_2$.

In Problems 9–12, find the modulus of the given complex number.

9. $(1 - i)^2$

10. $i(2 - i) - 4(1 + \frac{1}{4}i)$

11. $\dfrac{2i}{3 - 4i}$

12. $\dfrac{1 - 2i}{1 + i} + \dfrac{2 - i}{1 - i}$

In Problems 13 and 14, let $z = x + iy$. Express the given quantity in terms of x and y.

13. $|z - 1 - 3i|^2$

14. $|z + 5\bar{z}|$

In Problems 15 and 16, determine which of the given two complex numbers is closest to the origin. Which is closest to $1 + i$?

15. $10 + 8i,\ 11 - 6i$

16. $\frac{1}{2} - \frac{1}{4}i,\ \frac{2}{3} + \frac{1}{6}i$

In Problems 17–26, describe the set of points z in the complex plane that satisfy the given equation. Include a plot of the set of points in the complex plane.

17. $\text{Re}((1 + i)z - 1) = 0$

18. $[\text{Im}(i\bar{z})]^2 = 2$

19. $|z - i| = |z - 1|$

20. $\bar{z} = z^{-1}$

21. $\text{Im}(z^2) = 2$

22. $\text{Re}(z^2) = \left|\sqrt{3} - i\right|$

23. $|z - 1| = 1$

24. $|z - i| = 2\,|z - 1|$

25. $|z - 2| = \text{Re}(z)$

26. $|z| = \text{Re}(z)$

In Problems 27 and 28, establish the given simultaneous inequality.

27. If $|z| = 2$, then $8 \le |\,z + 6 + 8i\,| \le 12$.

28. If $|z| = 1$, then $2 \le \left|\,z^2 - 3\,\right| \le 4$.

29. Find an upper bound for the modulus of $3z^2 + 2z + 1$ if $|z| \le 1$.

30. Find an upper and lower bound for the reciprocal of the modulus of $z^4 - 5z^2 + 6$ if $|z| = 2$. [*Hint*: $z^4 - 5z^2 + 6 = (z^2 - 3)(z^2 - 2)$.]

In Problems 31 and 32, find a number z that satisfies the given equation.

31. $|z| - z = 2 + i$

32. $|z|^2 + 1 + 12i = 6z$

Focus on Concepts

33. (**a**) Draw the pair of points $z = a + ib$ and $\bar{z} = a - ib$ in the complex plane if

$$a > 0, b > 0; a > 0, b < 0; a < 0, b > 0; \text{and} a < 0,\ b < 0.$$

(**b**) In general, how would you describe geometrically the relationship between a complex number $z = a + ib$ and its conjugate $\bar{z} = a - ib$?

(**c**) Describe geometrically the relationship between $z = a + ib$ and $z_1 = -a + ib$.

34. How would you describe geometrically the relationship between a nonzero complex number $z = a + ib$ and its

(**a**) negative, $-z$?

(**b**) inverse, z^{-1}? [*Hint*: Reread Problem 33 and then recall $z^{-1} = \bar{z}/|z|^2$.]

35. Consider the complex numbers $z_1 = 4 + i$, $z_2 = -2 + i$, $z_3 = -2 - 2i$, $z_4 = 3 - 5i$.

(**a**) Use four different sketches to plot the four pairs of points z_1, iz_1; z_2, iz_2; z_3, iz_3; and z_4, iz_4.

(**b**) In general, how would you describe geometrically the effect of multiplying a complex number $z = x + iy$ by i? By $-i$?

36. Use Definition 1.1.2 to prove that $|z| = 0$ if and only if $z = 0$.

37. Under what circumstances does $|z_1 + z_2| = |z_1| + |z_2|$?

38. Using the complex variable z, find an equation of a circle in the complex plane of radius 5 centered at $3 - 6i$. [*Hint*: Use (5).]

39. Describe the set of points z in the complex plane that satisfy $z = \cos\theta + i\sin\theta$, where θ is measured in radians from the positive x-axis.

40. Using the complex variable z, find an equation of an ellipse in the complex plane with foci $-2 + i$ and $2 + i$ whose major axis is 8 units long.

41. Suppose $z = x + iy$. In (6) of Section 1.1 we saw that x and y could be expressed in terms of z and $\bar{z}$. Use these results to express the following Cartesian equations in complex form.

(**a**) $x = 3$ (**b**) $y = 10$

(**c**) $y = x$ (**d**) $x + 2y = 8$

42. Using complex notation, find a parametric equation of the line segment between any two distinct complex numbers z_1 and z_2 in the complex plane.

43. Suppose z_1, z_2, and z_3 are three distinct points in the complex plane and k is a real number. Interpret $z_3 - z_2 = k(z_2 - z_1)$ geometrically.

44. Suppose $z_1 \neq z_2$. Interpret $\text{Re}(z_1\bar{z}_2) = 0$ geometrically in terms of vectors z_1 and z_2.

45. Suppose $w = \bar{z}/z$. Without doing any calculations, explain why $|w| = 1$.

46. Without doing any calculations, explain why the inequalities $|\text{Re}(z)| \leq |z|$ and $|\text{Im}(z)| \leq |z|$ hold for all complex numbers z.

47. Show that

(**a**) $|z| = |-z|$ (**b**) $|z| = |\bar{z}|$.

48. For any two complex numbers z_1 and z_2, show that

$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2).$$

49. In this problem we will start you out in the proof of the first property $|z_1 z_2| = |z_1|\,|z_2|$ in (3). By the first result in (2) we can write $|z_1 z_2|^2 = (z_1 z_2)\overline{(z_1 z_2)}$. Now use the first property in (2) of Section 1.1 to continue the proof.

50. In this problem we guide you through an analytical proof of the triangle inequality (6).

Since $|z_1 + z_2|$ and $|z_1| + |z_2|$ are positive real numbers, we have

$|z_1 + z_2| \leq |z_1| + |z_2|$ if and only if $|z_1 + z_2|^2 \leq (|z_1| + |z_2|)^2$. Thus, it suffices to show that $|z_1 + z_2|^2 \leq (|z_1| + |z_2|)^2$.

(**a**) Explain why $|z_1 + z_2|^2 = |z_1|^2 + 2\text{Re}(z_1\bar{z}_2) + |z_2|^2$.

(**b**) Explain why $(|z_1| + |z_2|)^2 = |z_1|^2 + 2\,|z_1\bar{z}_2| + |z_2|^2$.

(**c**) Use parts (a) and (b) along with the results in Problem 46 to derive (6).